

- In general, a number expressed in base- $r$  system has coefficients multiplied by powers of  $r$ .

$$a_n \cdot r^n + a_{n-1} \cdot r^{n-1} + \dots + a_2 \cdot r^2 + a_1 \cdot r^1 + a_0 \cdot r^0 + a_{-1} \cdot r^{-1} + a_{-2} \cdot r^{-2} + \dots + a_{-m} \cdot r^{-m}$$

- The coefficients  $a_i$  range in values from 0 to  $r-1$ .
- An example of a base-5 number is:  $(4021)_5 = 4 \times 5^3 + 0 \times 5^2 + 2 \times 5^1 + 1 \times 5^0 + 2 \times 5^{-1} = (511.4)_{10}$ .
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- Note that coefficient values for base 5 can only be 0, 1, 2, 3 and 4.
- The letters of the alphabet are used to supplement the ten decimal digits when the base of the number is greater than 10.
- For example, in the hexadecimal (base 16) number system, the first ten digits are borrowed from the decimal system. The letters A, B, C, D, E and F are used for digits 10, 11, 12, 13, 14 and 15 respectively.
- An example of a hexadecimal number is:  $(B65F)_{16} = 11 \times 16^3 + 6 \times 16^2 + 5 \times 16^1 + 15 \times 16^0 = (46687)_{10}$ .

| Numbers with different bases |                 |                |                       |
|------------------------------|-----------------|----------------|-----------------------|
| Decimal<br>(base 10)         | Binary (base 2) | Octal (base 8) | Hexadecimal (base 16) |
| 00                           | 0000            | 00             | 0                     |
| 01                           | 0001            | 01             | 1                     |
| 02                           | 0010            | 02             | 2                     |
| 03                           | 0011            | 03             | 3                     |
| 04                           | 0100            | 04             | 4                     |
| 05                           | 0101            | 05             | 5                     |
| 06                           | 0110            | 06             | 6                     |
| 07                           | 0111            | 07             | 7                     |
| 08                           | 1000            | 08             | 8                     |
| 09                           | 1001            | 09             | 9                     |
| 10                           | 1010            |                |                       |